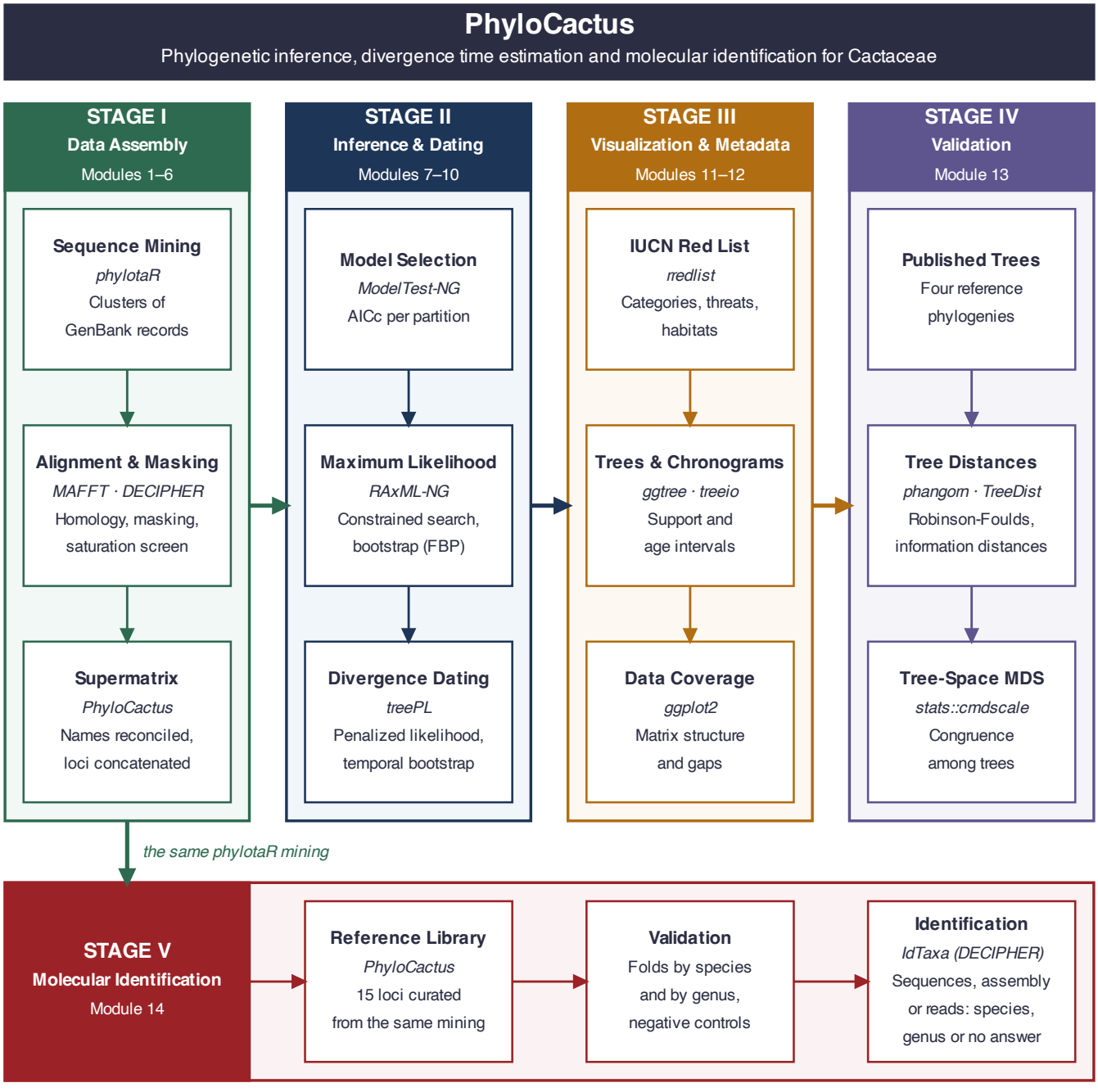
PhyloCactus
Phylogenetic inference, divergence time estimation and molecular identification for Cactaceae
STAGE I
Data Assembly
Modules 1–6
Sequence Mining
phylotaR
Clusters of GenBank records
Alignment & Masking
MAFFT · DECIPHER
Homology, masking, saturation screen
Supermatrix
PhyloCactus
Names reconciled, loci concatenated
STAGE II
Inference & Dating
Modules 7–10
Model Selection
ModelTest-NG
AICc per partition
Maximum Likelihood
RAxML-NG
Constrained search, bootstrap (FBP)
Divergence Dating
treePL
Penalized likelihood, temporal bootstrap
STAGE III
Visualization & Metadata
Modules 11–12
IUCN Red List
rredlist
Categories, threats, habitats
Trees & Chronograms
ggtree · treeio
Support and age intervals
Data Coverage
ggplot2
Matrix structure and gaps
STAGE IV
Validation
Module 13
Published Trees
Four reference phylogenies
Tree Distances
phangorn · TreeDist
Robinson-Foulds, information distances
Tree-Space MDS
stats::cmdscale
Congruence among trees
the same phylotaR mining
STAGE V
Molecular Identification
Module 14
Reference Library
PhyloCactus
15 loci curated from the same mining
Validation
Folds by species and by genus, negative controls
Identification
IdTaxa (DECIPHER)
Sequences, assembly or reads: species, genus or no answer